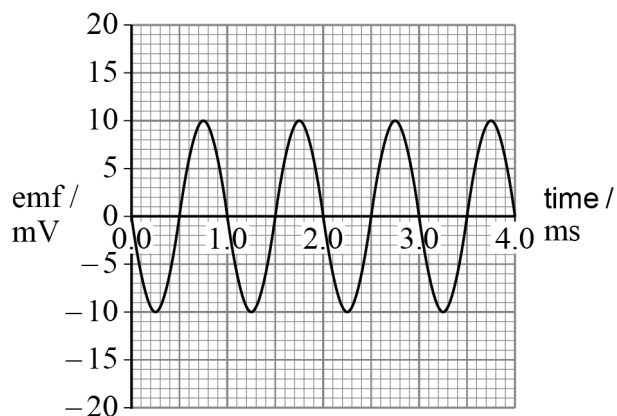


3

1

A coil of wire rotates at a constant angular speed ω in a uniform magnetic field. The graph shows the variation with time of the emf induced in the coil.



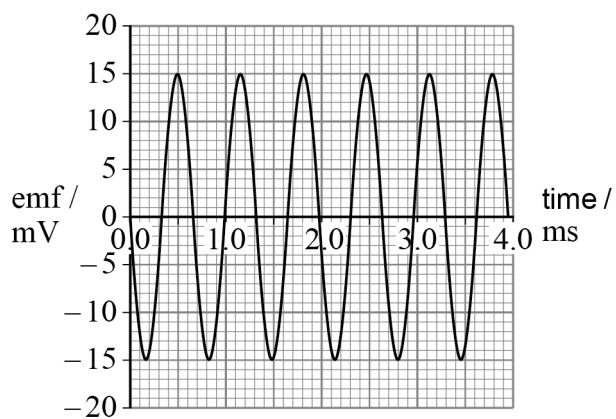
The angular speed of the coil is changed to $\frac{3\omega}{2}$.

Which graph shows the new variation with time of the emf induced in the coil?

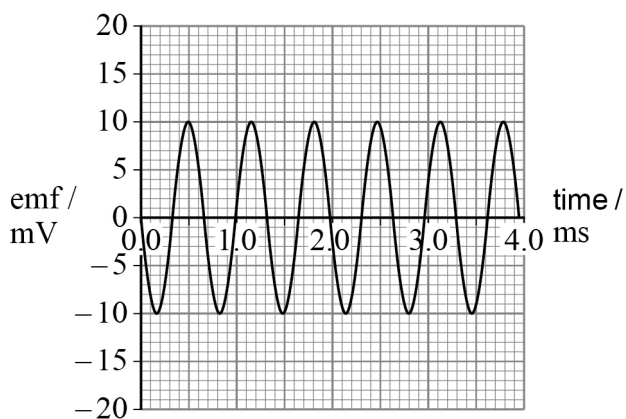
[1 mark]



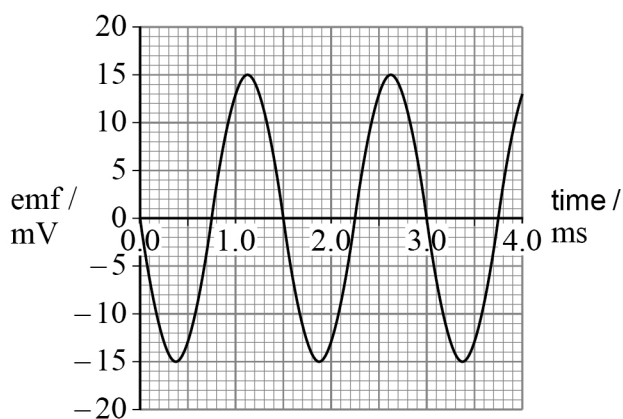
A



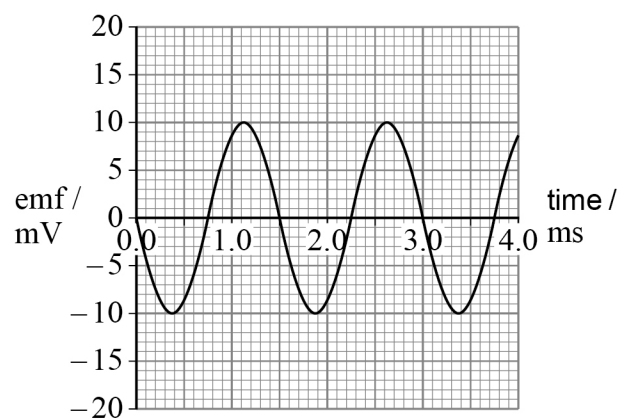
B



C



D



A



B



C



D

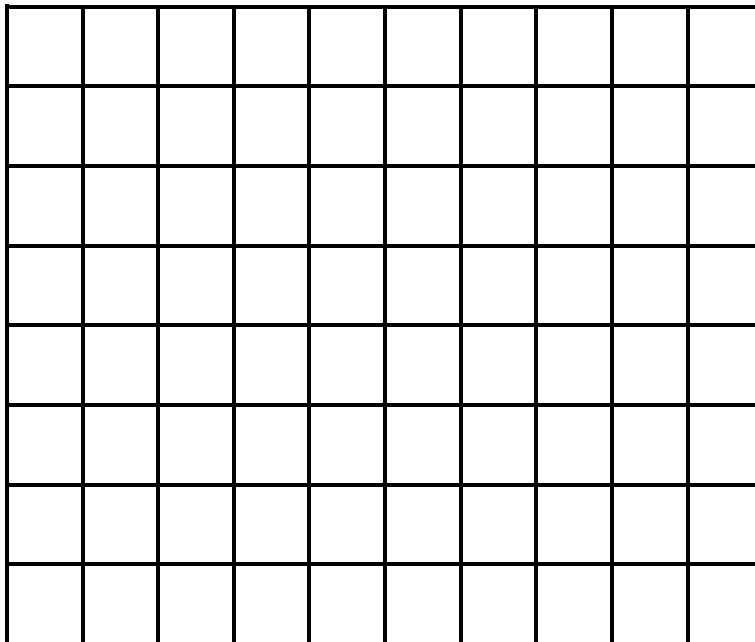


Turn over for the next question

Turn over ►



Questions **32** and **33** are about using an oscilloscope to show an ac waveform from an ac source. The oscilloscope screen is represented below.



3**4**

Which of these does **not** increase the efficiency of a transformer?

[1 mark]

A Increasing the diameter of the wire of the secondary coil.

☐

B Overlapping the primary coil with the secondary coil.

☐

C Using a steel core instead of an iron core.

☐

D Using a core of thin insulated metal sheets.

☐

3 5 A **step-up** transformer is 100% efficient.

What is a possible set of values for the transformer?

[1 mark]

	N_p	V_p	N_s	V_s
A	300	150	100	50
B	150	300	100	50
C	100	150	300	50
D	100	50	300	150

☐☐☐☐

[illegible]

[illegible]

[illegible]

[illegible]

[illegible]

Figure 10 shows apparatus used to investigate the motion of a trolley down a ramp. The trolley has a magnet attached to it. When released from point **X**, the trolley moves down the ramp at a **constant speed**.

An emf is induced in a circular coil of wire as the trolley and magnet pass under the coil. The coil is connected to a datalogger that records the induced emf.

Figure 10

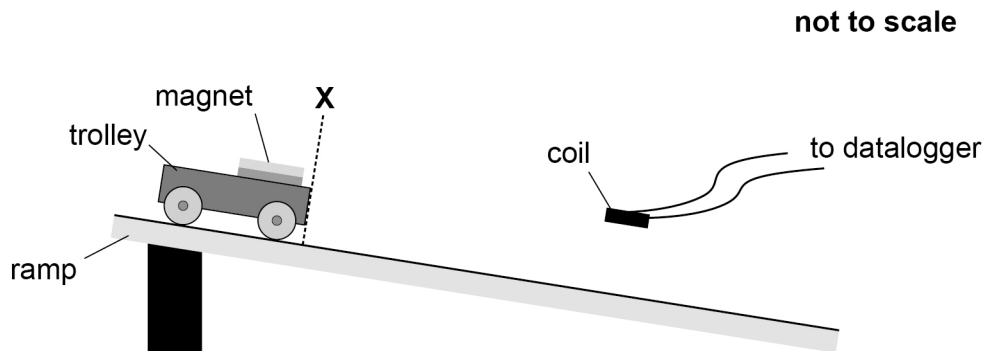
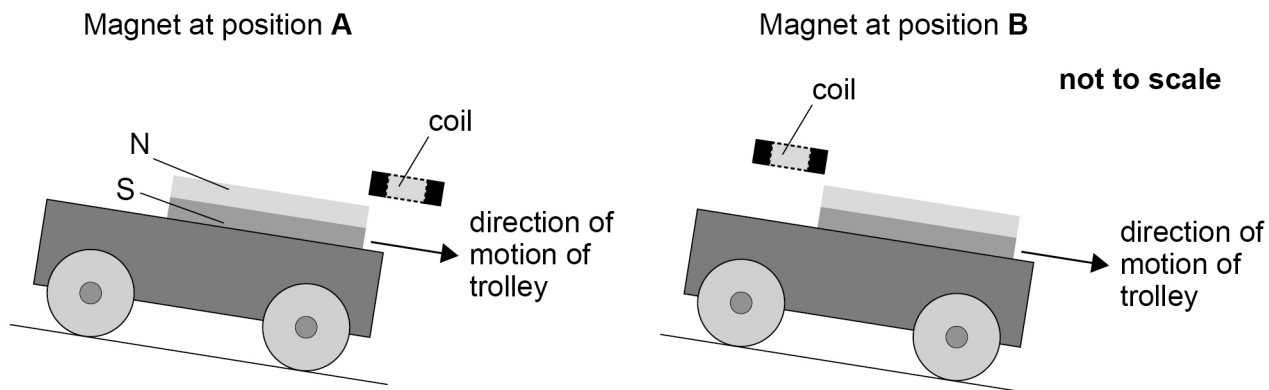


Figure 11 shows expanded views of the magnet passing under the coil. Two positions are shown as the magnet moves down the ramp.

Position **A** is when the front edge of the magnet reaches the left-hand side of the coil. Position **B** is when the trailing edge of the magnet leaves the right-hand side of the coil.

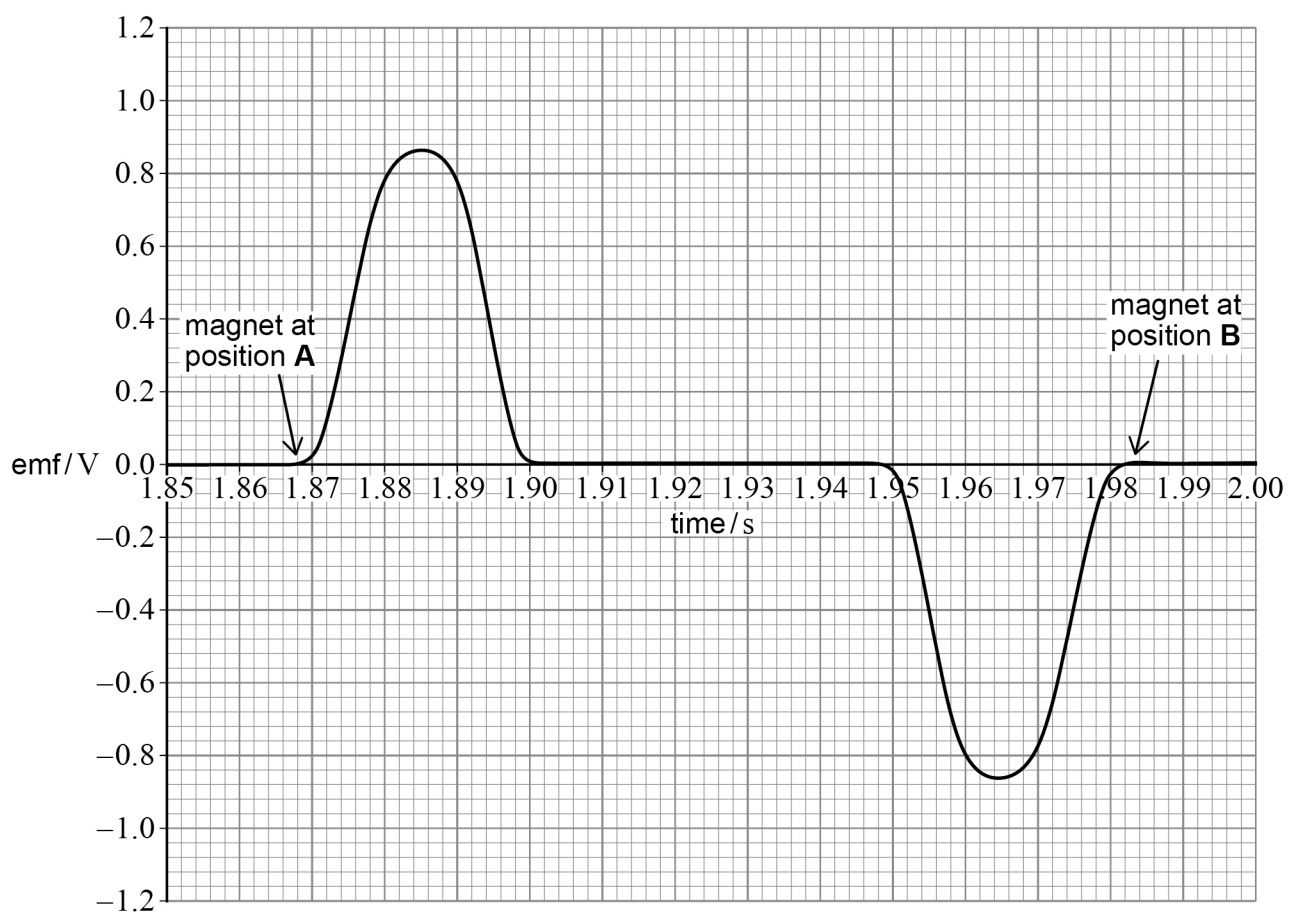
Figure 11



Question 5 continues on the next page

Figure 12 shows how the induced emf in the coil varies with time after the trolley is released from point **X**.

Figure 12



0 5 . 1 Determine the maximum rate of change of flux linkage in the coil.

[1 mark]

maximum rate of change of flux linkage = _____ Wb s^{-1}



0 5 . 2 Explain the shape of the graph in **Figure 12**.

[3 marks]

0 5 . 3 The length of the magnet is 4.0 cm and the diameter of the coil is 1.5 cm.

Estimate, using **Figure 12**, the speed of the trolley between positions **A** and **B**.

[2 marks]

speed = _____ m s⁻¹

Question 5 continues on the next page

Turn over ►



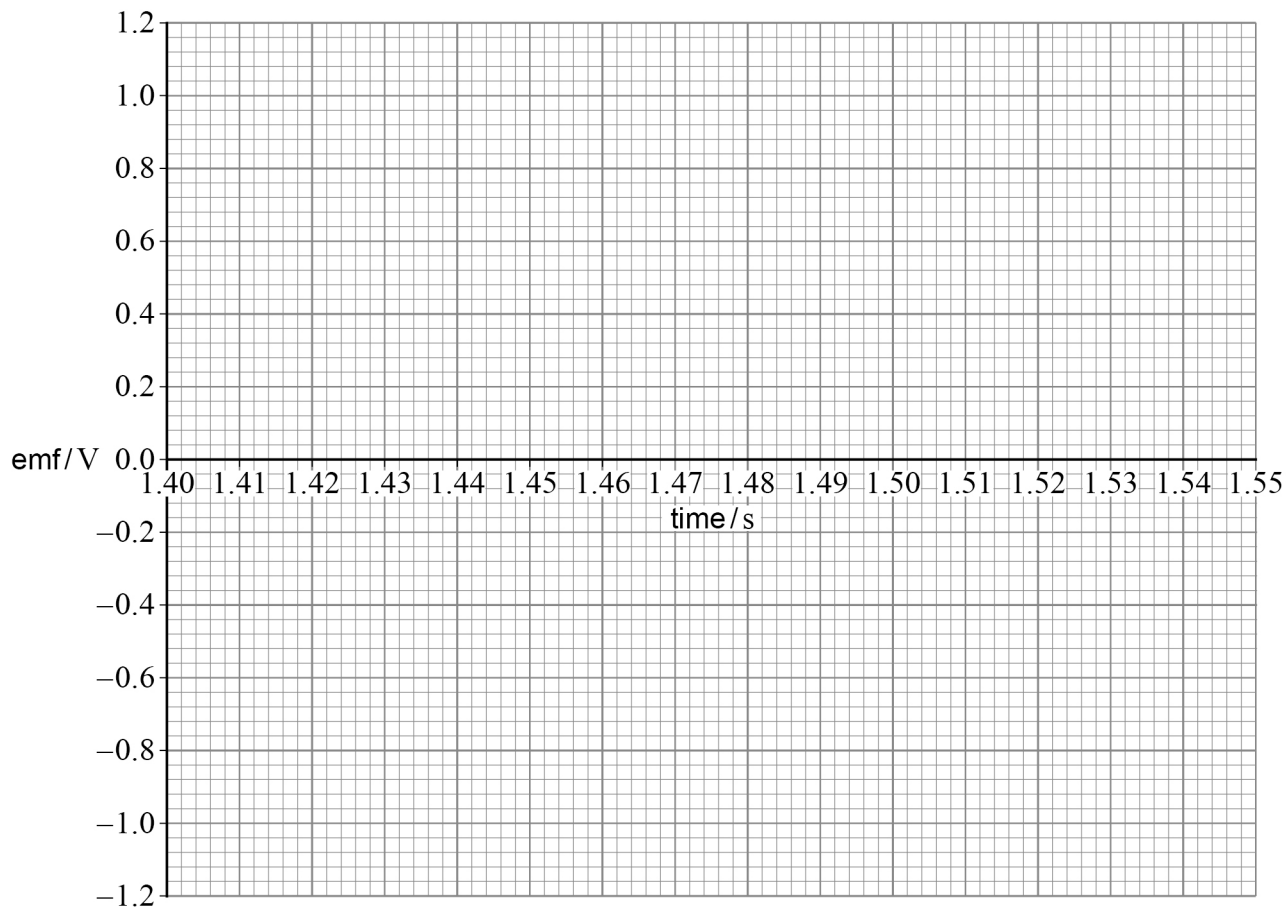
0 5 . 4

The angle of the ramp is increased and the trolley is again released from point **X**. The trolley now accelerates down the ramp and arrives at position **A** at 1.42 s.

Sketch, on **Figure 13**, the new variation of induced emf with time as the magnet travels from position **A** to position **B**.

[4 marks]

Figure 13



END OF SECTION A



Section B

Each of the questions in this section is followed by four responses, **A**, **B**, **C** and **D**.

For each question select the best response.

Only **one** answer per question is allowed.

For each question, completely fill in the circle alongside the appropriate answer.

CORRECT METHOD



WRONG METHODS



If you want to change your answer you must cross out your original answer as shown.



If you wish to return to an answer previously crossed out, ring the answer you now wish to select as shown.



You may do your working in the blank space around each question but this will not be marked.
Do **not** use additional sheets for this working.

3**1**

Lenz's law is an example of the conservation of

[1 mark]

A charge.



B energy.



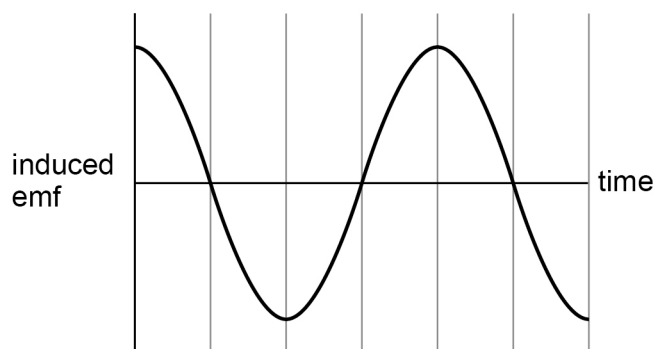
C flux linkage.



D momentum.



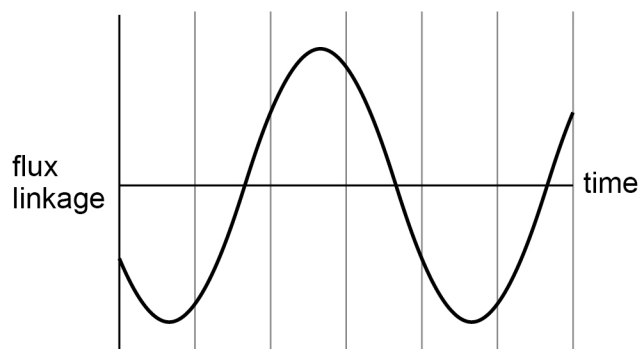
The graph shows the variation of induced emf with time for a coil rotating in a magnetic field.



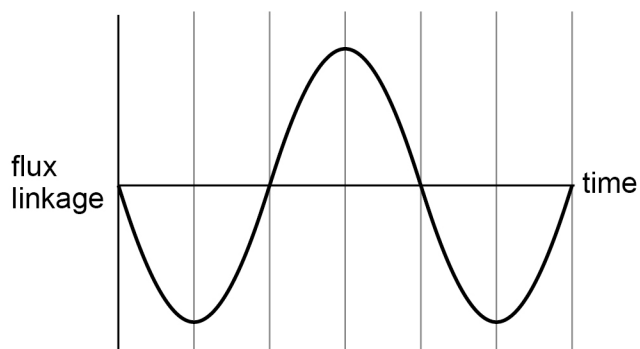
Which graph shows the variation of flux linkage in the coil with time?

[1 mark]

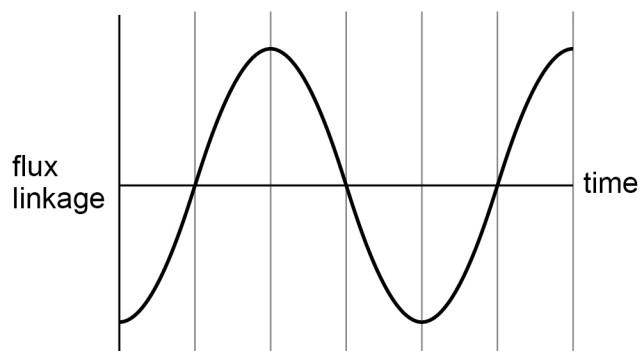
A



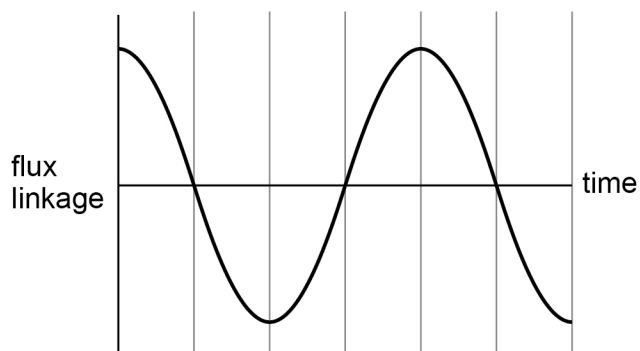
B



C



D



A

☐

B

☐

C

☐

D

☐


Do not write	Do not write
--------------	--------------

An ideal transformer has 300 turns in the primary coil and 600 turns in the secondary coil.

The primary coil is connected to an ac source that has a peak potential difference of 10 V and a frequency of 20 Hz.

What is the peak potential difference and the frequency of the output from the secondary coil?

[1 mark]

	Peak potential difference / V	Frequency / Hz
A	5	20
B	5	40
C	20	20
D	20	40

☐☐☐☐

[illegible]

[illegible]

[illegible]

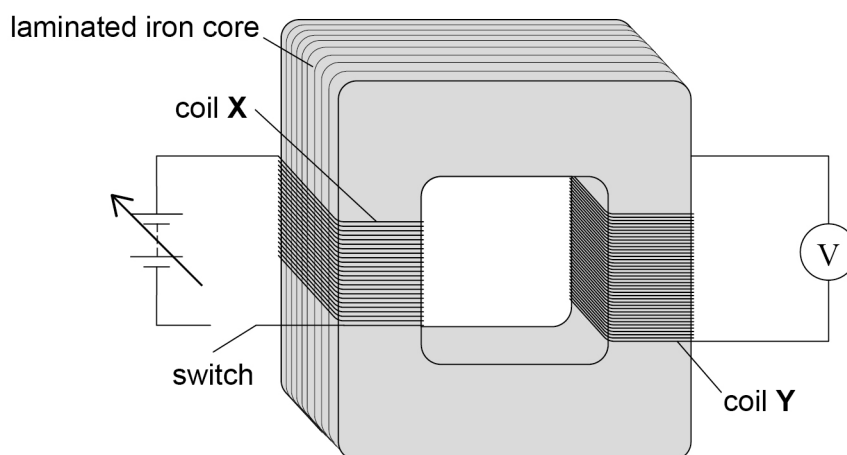
[2 marks]

fundamental (base) units = _____

Figure 4 shows the arrangement that a student uses to investigate electromagnetic induction.

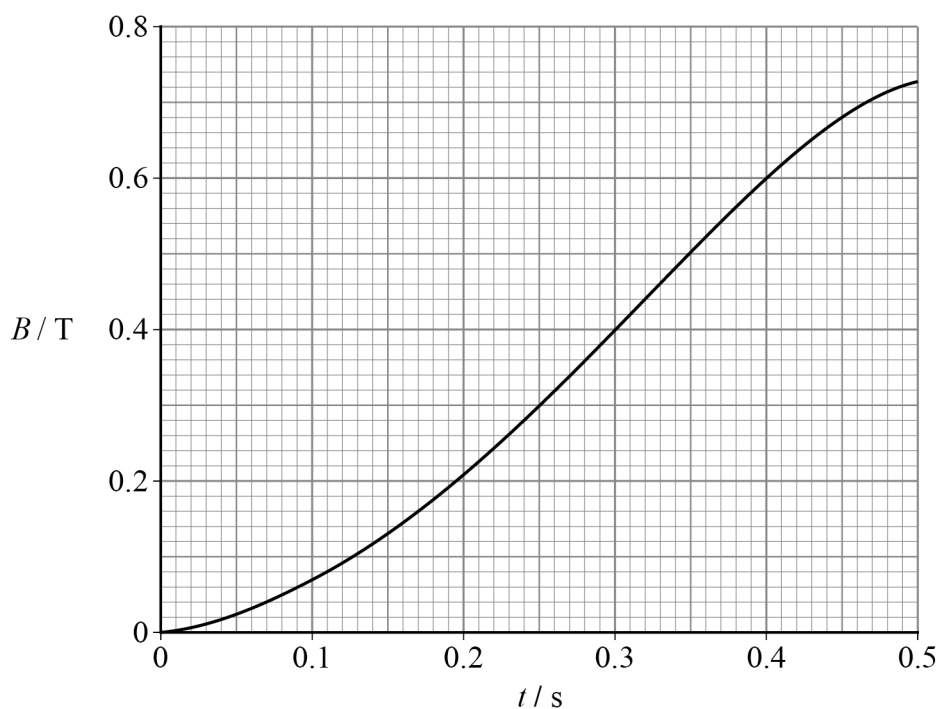
The voltmeter connected across coil **Y** can measure both positive and negative potential differences.

Figure 4



The student closes the switch and gradually adjusts the variable resistor so that the magnetic flux density B in the core varies with time t as shown in **Figure 5**.

Figure 5



0 5 . 2

The voltmeter measures ε , the induced emf across coil **Y**.

Show that

$$\varepsilon = ANm$$

where

A is the cross-sectional area of the core

N is the number of turns on coil **Y**

m is the gradient of the line shown in **Figure 5**.

[2 marks]

Question 5 continues on the next page

Turn over ►



Estimate the maximum value of ε during the 0.5 s shown in **Figure 5**.

$$N = 200$$

[3 marks]

maximum value of $\varepsilon =$ V

The student opens the switch and the current in coil **X** decreases to zero in a very short time.

Describe what happens to the reading on the voltmeter during this time. Go on to explain why this happens.

[3 marks]

[illegible]

0 5 . 5

The arrangement shown in **Figure 4** can be used as a transformer when coil **X** is connected to an alternating current supply.

Explain the reason for using a laminated core.

[2 marks]

12

Turn over for the next question

Turn over ►



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0

7

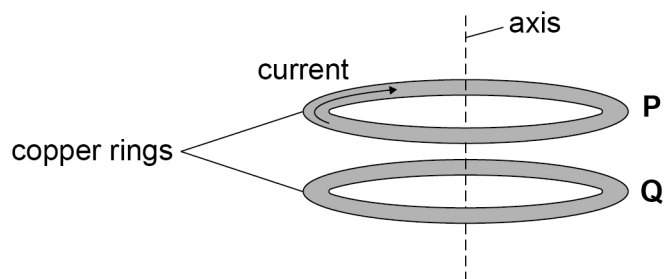
Which row identifies a vector and a scalar?

[1 mark]

	Vector	Scalar
A	centripetal force	gravitational field strength
B	gravitational potential	electric potential energy
C	centripetal acceleration	capacitance
D	time constant	magnetic flux linkage

☐☐☐☐

Two identical horizontal copper rings **P** and **Q** are fixed on a common axis. **P** is vertically above **Q**. Initially, there is a clockwise current in **P** and no current in **Q**.



The clockwise current in **P** is reduced at a constant rate.

This reduction results in

[1 mark]

A a clockwise current in **Q**.



B an anticlockwise current in **Q**.



C an alternating current in **Q**.



D no current in **Q**.



Turn over for the next question



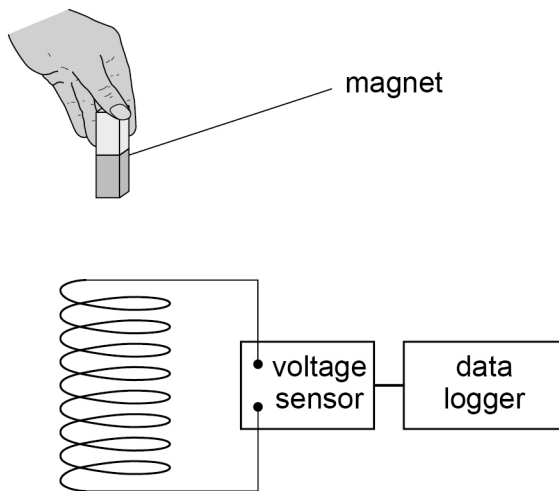
Do not write

A student does an experiment to investigate electromagnetic induction.

She drops a magnet vertically through a coil of wire.

The coil is connected to a voltage sensor and data logger, as shown in **Figure 9**.

Figure 9



06.1

Explain why an emf is induced as the magnet falls towards the coil.

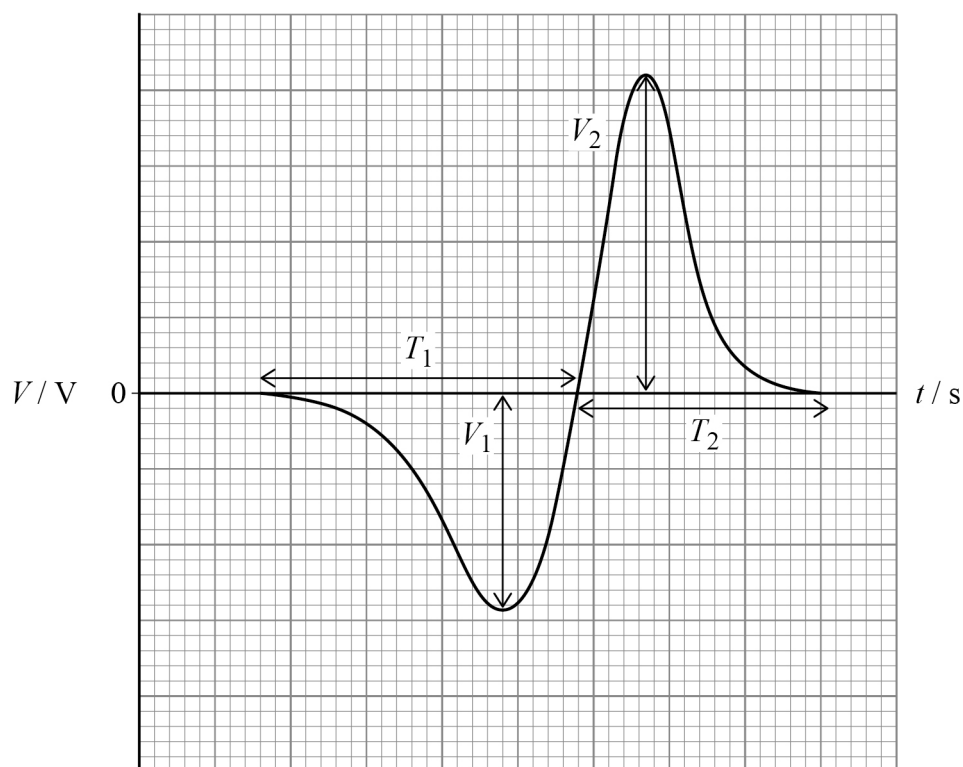
[2 marks]

Question 6 continues on the next page



Figure 10 shows the variation of the induced voltage V with time t after the magnet is released.

Figure 10



The durations T_1 and T_2 and maximum values V_1 and V_2 of two peaks are labelled.

0 6 . 2

Explain why V_1 and V_2 have opposite signs.

[2 marks]



0 6 . 3

Explain why the magnitude of V_1 is less than the magnitude of V_2 .

[2 marks]

0 6 . 4

Explain why T_1 is greater than T_2 .

[2 marks]

8

END OF SECTION A

Turn over ►



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outside the
box*

**DO NOT WRITE ON THIS PAGE
ANSWER IN THE SPACES PROVIDED**



Section B

Each of the questions in this section is followed by four responses, **A**, **B**, **C** and **D**.

For each question select the best response.

Only **one** answer per question is allowed.

For each question, completely fill in the circle alongside the appropriate answer.

CORRECT METHOD



WRONG METHODS



If you want to change your answer you must cross out your original answer as shown.

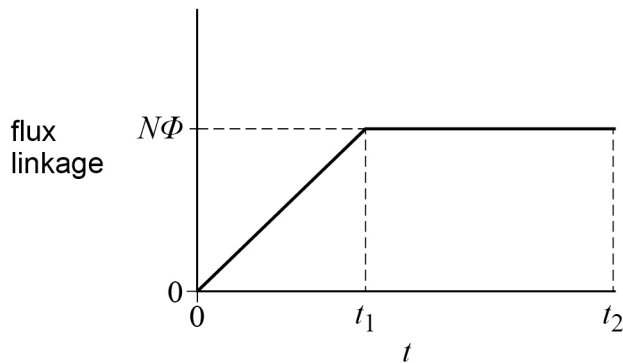


If you wish to return to an answer previously crossed out, ring the answer you now wish to select as shown.



You may do your working in the blank space around each question but this will not be marked.
Do **not** use additional pages for this working.

The graph shows how the flux linkage in a coil varies with time t .



What is the average emf induced in the coil between $t = 0$ and $t = t_2$?

[1 mark]

A $\frac{N\Phi}{t_1}$

☐

B $\frac{N\Phi}{t_2}$

☐

C $N\Phi\left(t_2 - \frac{t_1}{2}\right)$

☐

D $\frac{N\Phi}{2t_1}$

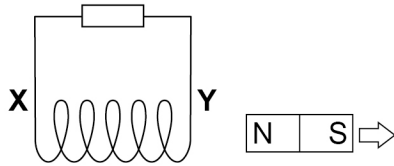
☐

A bar magnet is moved near a coil **XY** in the direction shown by the arrows.
Two of the actions shown in the diagrams will induce an N-pole at Y.

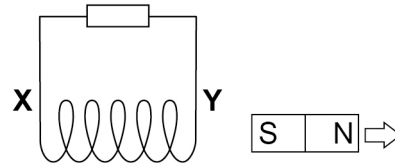
Which actions will induce an N-pole at **Y**?

[1 mark]

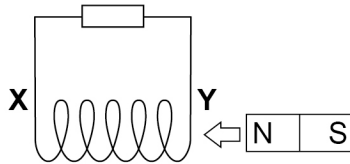
1



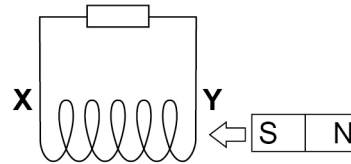
2



3



4



A 1 and 3

☐

B 1 and 4

☐

C 2 and 3

☐

D 2 and 4

☐

[illegible]

[illegible]

This image shows a single sheet of white paper with horizontal ruling lines. The lines are evenly spaced and run across the width of the page. There is no handwriting or other markings on the paper.

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The root mean square (rms) voltage across a resistor is 125 V and the peak current in the resistor is 1.7 A.

What is the mean power dissipated in the resistor?

[1 mark]

A 106 W

☐

B 150 W

☐

C 213 W

☐

D 301 W

☐

Turn over ►



Do not write

A transformer is used to produce a higher voltage from a 5.0 V rms supply. The transformer has a primary coil of 1000 turns and a secondary coil of 12 000 turns. All of the flux from the primary coil links with the secondary coil.

The rms current in the secondary coil is 0.15 A and the transformer is 90% efficient.

What is the rms current in the primary coil?

[1 mark]

A 1.6 A

☐

B 1.8 A

☐

C 2.0 A

☐

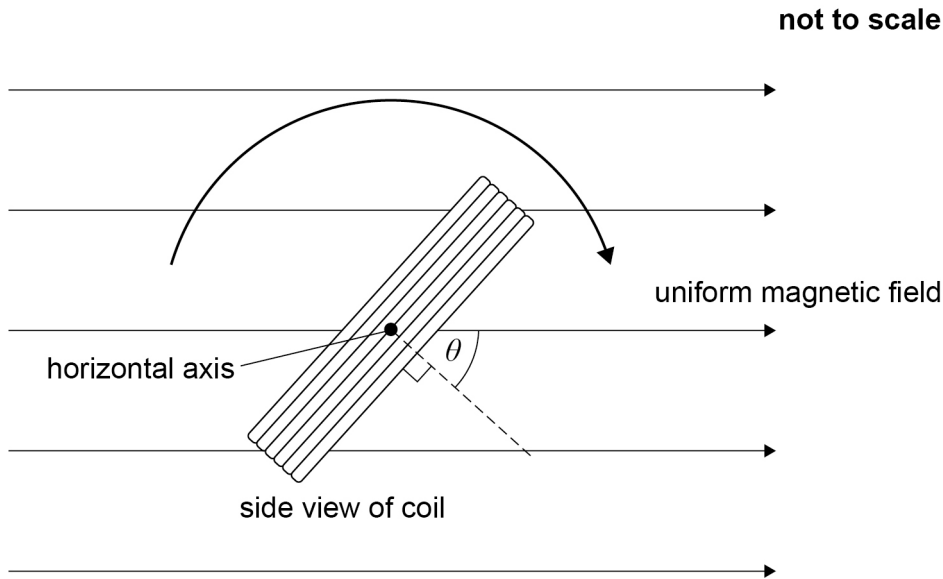
D 2.3 A

☐

0 6

A coil of wire in a uniform magnetic field rotates about its horizontal axis at a constant frequency f . **Figure 13** shows the side view of the coil at one instant.

Figure 13



The coil has an area A of $7.5 \times 10^{-4} \text{ m}^2$ and N turns.
The field has a magnetic flux density of $4.2 \times 10^{-2} \text{ T}$.
 $\theta = 51^\circ$

0 6 . 1

For the position shown in **Figure 13**, the flux linkage is $2.4 \times 10^{-3} \text{ Wb}$ for the coil.

Calculate N .

[1 mark]

$N =$ _____

0 6 . 2

Explain why an emf is induced in the coil.
Refer to Faraday's law.

[2 marks]

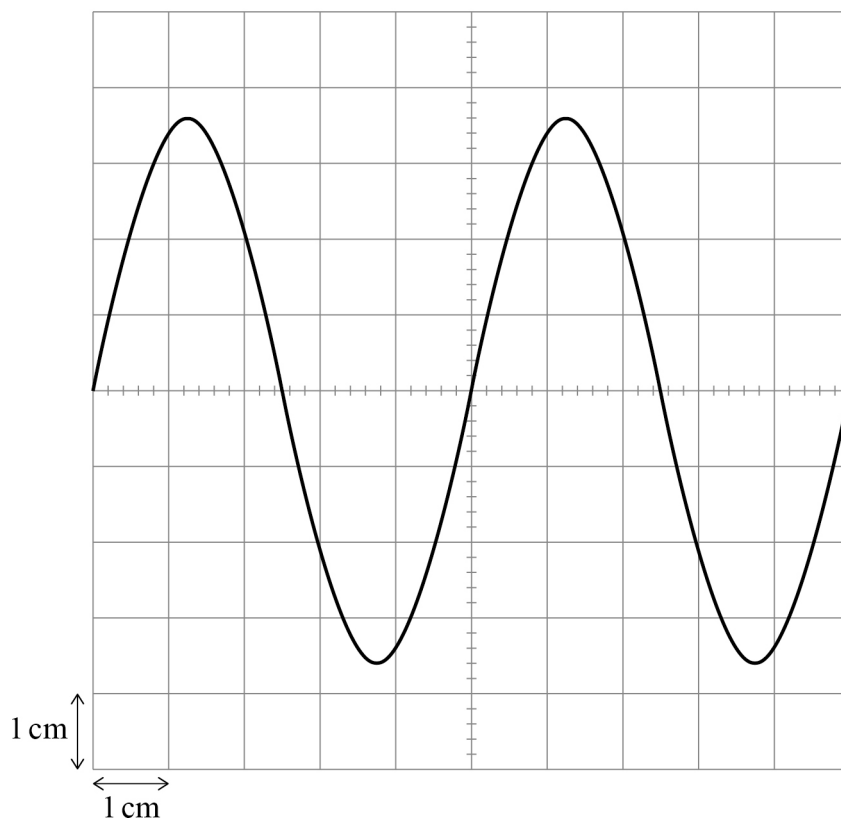
Question 6 continues on the next page

Turn over ►



Figure 14 shows a trace on an oscilloscope displaying the emf induced in the rotating coil.

Figure 14



0 6 . 3

Annotate **Figure 14** with an **X** to show a point on the trace that corresponds to the position of the coil shown in **Figure 13**. Support your answer with a suitable calculation.

[2 marks]



The induced emf has a root mean square value of 1.27 V.

0 6 . 4

The frequency f of rotation of the coil is 50.0 Hz.

Determine the settings for the time-base and the y -gain on the oscilloscope.

[4 marks]

time-base = _____ ms cm⁻¹

y -gain = _____ V cm⁻¹

Question 6 continues on the next page



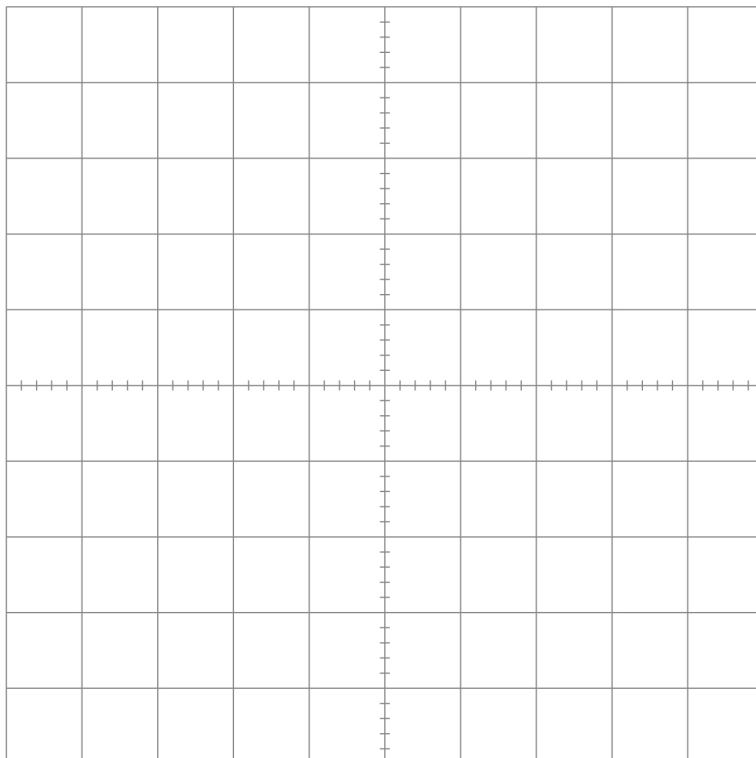
0 6 . 5

The coil now rotates with a frequency of 25 Hz. The settings on the oscilloscope are the same settings as those used to make the trace in **Figure 14**.

Sketch the new trace on **Figure 15**.

[2 marks]

Figure 15



Don't write

Eddy currents are produced by the changing magnetic field in the laminated iron core of a transformer.

Which statement about eddy currents is correct?

[1 mark]

- A** They are in the same plane as the laminations of the core. ☐
- B** They change direction when the magnetic field changes direction. ☐
- C** Their size depends on the resistivity of the material of the core. ☐
- D** They do not require an induced emf in the core to be produced. ☐

Turn over for the next question





A single-turn rectangular coil of area A rotates at frequency f in a uniform magnetic field. The peak value of the induced emf is ε .

A second single-turn rectangular coil rotates in the same magnetic field and has the same peak induced emf ε .

Which row shows the area and frequency of rotation of the second coil?

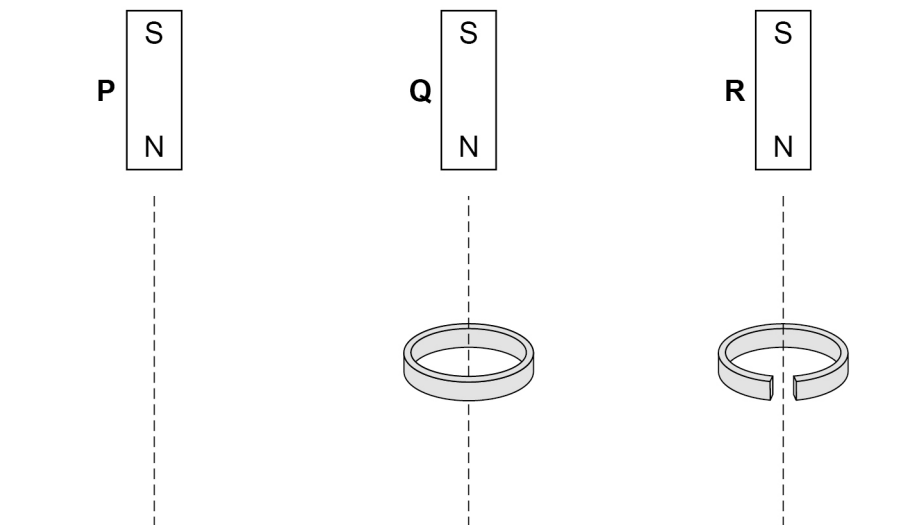
[1 mark]

	Area of coil	Frequency of rotation	
A	$4A$	$0.5f$	☐
B	$2A$	$2f$	☐
C	$2A$	$0.5f$	☐
D	$0.25A$	$2f$	☐



Don't write

The diagram shows three identical magnets **P**, **Q** and **R**.



The magnets are released from the same height, at the same time, and fall to the ground.

P falls directly to the ground.

Q falls through the centre of a thick conducting ring.

R falls through an identical ring that has a gap cut in it.

Which gives the order that the magnets reach the ground?

[1 mark]

A **P** and **R** arrive at the same time, followed by **Q**.

☐

B **P** and **Q** arrive at the same time, followed by **R**.

☐

C **P** arrives first, followed by **Q** and then **R**.

☐

D **P**, **Q** and **R** arrive at the same time.

☐

2	0
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An ac supply transfers energy to a resistor. An rms current of 8 A and a direct current I transfer energy at the same rate in the resistor.

What is I ?

[1 mark]

A $2\sqrt{2}$ A

☐

B 4 A

☐

C $4\sqrt{2}$ A

☐

D 8 A

☐

Do not write

2

1

A transformer has 2000 turns on the primary coil and 900 turns on the secondary coil. A 230 V ac supply produces a current of 0.10 A in the primary coil. The current in the secondary coil is 0.21 A.

All of the flux produced by the primary coil links the secondary coil.

What is the efficiency of the transformer?

[1 mark]

A 45%

☐

B 48%

☐

C 95%

☐

D 100%

☐

2

2

A transformer with 100% efficiency has eight times more turns in the primary coil than it has in the secondary coil.

When in use:

- the mean input power is 10 kW
- the rms current in the secondary coil is 25 A.

What is the rms voltage across the primary coil?

[1 mark]

A 50 V

☐

B 400 V

☐

C 3200 V

☐

D 31 300 V

☐

2**3**

A resistor dissipates a mean power of 48 W and has an rms voltage of 12 V ac across it.

What is the peak current in the resistor?

[1 mark]

A 1.4 A

☐

B 2.8 A

☐

C 4.0 A

☐

D 5.7 A

☐

[illegible]

[illegible]

[illegible]

[illegible]

5

[6 marks]

--	--